1970 – OR The Beginning of Time

Introduction

- Anybody knows what's happened then?
- Have you ever executed a piece of code such as:
  - // C program calling time()
  - // Shell calling time
- OR

History

- Why was Unix written, built & how did it arise?
- Other alternatives
  - Multics &
  - the best programmer's OS's

Insert some here

- Explain "time()"'s rv
- Brief introduction to 70's state of comp progress
  - * Machines, TTY, PF lang. + SPE Lect
- Brief introduction to KT
  - * Show a picture of him
    - * it is an el engineer
    - * it is a big chess fan
- Brief introduction of DR.
- The other guys (?)
  - * Mcilroy, Bourne, Bourne ...
- How everything began
  - * The game made by KT.
  - * The 3-week study of KT.
  - * The C prog. language
  - * The PDP-11 rewrite
- The 'result'

Applications and Consequences

- Short story about young guy named Linus who has a lot of Unix & and how his lack of money + good experience Finland's from USA ID
  - * e.g. The begging of FREAK.

Last words.

- What Unix today??
  - * No mainframe
  - * No smartphones
  - * No [a-lot-of-things]
- The heroes:
  - Ken Th. → You know it: B!!!, Unix; Go
  - Dennis Ritchie: C & Unix
  - McIlroy → "the smartest among all of us" – Ken
  - Bourne → BASH
  - Br. Kernighan → The C programming language" Book; AWK

  Lot, more for a half sentence.
- Knowledge!
- Questions

  focus

Q: Why did you choose this topic and for example the creation of C (its value has long overtaken Unix) or Linux (its contr. is still unlimited)

A: About C

- C was a language developed for the purpose of better implementation of Unix
  - → Unix's success C granted its place as a language in which complex low-level systems could be built.

About Linux

- Linux wouldn't even exist without Unix.

Focus on this!

- Make it easy for them to understand the core concepts: kernel, vm, processes, fs, etc.
- Make them love it!
- Make them want to try it themselves.
  - Bonus: Try to demo a real working version of the code
    - :: Provide some scripts for them to build it themselves
    - :: Encourage them to add features
      - List some possibilities.

The Deep Stuff

- Design, Philosophy, Arch.
- Implementation details

Note that they don't use gcc; No POSIX → they invent it! Therefor the strange func. param. syntax (it's still valid)

How?

- Explain bootstrapping's basic idea &
- Then show the code